

Exome sequencing identifies potential variants linked to cardiovascular disease in Caldas, Colombia: insights from the ORIGEN COLOMBIAN genome project

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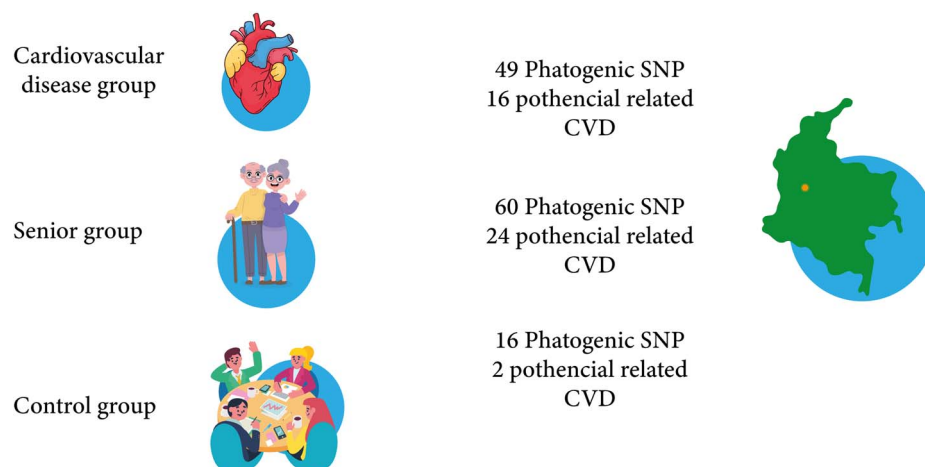
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Abstract

The ORIGEN Project identified, for the first time, variants potentially associated with cardiovascular disease (CVD) in individuals from the Caldas region of Colombia. These findings provide novel insights into the genetic architecture of CVD in an underrepresented population and contribute to closing the knowledge gap in Latin American genomics research. Peripheral blood samples were collected for DNA extraction and exome sequencing using the Oxford Nanopore® platform. Bioinformatics analysis was performed at the variables and their relationships were described. Statistical significance was demonstrated with a p-value less than 0.05. The study involved 250 individuals divided into three groups: individuals diagnosed with cardiovascular disease (CVD), older adults without a known CVD diagnosis, and a healthy control group. The most common diagnoses in the CVD group were hypertension (29%), acute myocardial infarction (27%), and heart failure (5%). Whole exome sequencing revealed six single nucleotide polymorphisms (SNPs) potentially associated with CVD in the genes *HFE*, *PERM1*, and *FBN1*. The number of pathogenic variants was significantly higher in the older adult group (60 variants), the CVD group (49 variants), and the control group (16 variants). The identification of pathogenic variants potentially associated with CVD and aging in our population opens new avenues for the advancement of precision medicine in the region and the country.

Graphical Abstract

EXOME SEQUENCING IDENTIFIES POTENTIAL VARIANTS LINKED TO CARDIOVASCULAR DISEASE IN CALDAS, COLOMBIA: INSIGHTS FROM THE ORIGEN COLOMBIAN GENOME PROJECT.



Keywords: Cardiovascular risk; Genomics; Exome; Single nucleotide polymorphism (SNP); Precision medicine

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Introduction

Cardiovascular diseases are the leading cause of death worldwide and impose a significant burden on individuals, families, and healthcare systems. According to the Colombian Health Situation Analysis [1], between 2005 and 2021, circulatory system diseases (CSD) were the leading cause of death in the general population. In 2021, CSD resulted in 151.94 deaths per 100 000 inhabitants per year, representing 26.01% (94432) of all deaths, and 20.5% (1109159) of all potential years of life lost (PYLL). This burden could be mitigated through enhanced preventive strategies and personalized treatments tailored to individual genetic profiles and lifestyles. This approach to healthcare lies at the core of what is now known as precision medicine [2].

The risk of ischemic heart disease is influenced not only by environmental and lifestyle factors, but also by individuals' genetic characteristics. This highlights the importance of understanding the genetic basis of heart disease [3]. In this sense, characterizing a person's risk of developing a disease can guide treatment and, if possible, prevent disease onset, delay its progression, or reduce its severity. Genomic information can support prevention efforts across different stages of life [4].

In Colombia, efforts in this area have been fragmented and limited in terms of genomic scope and population representativeness [5–8]. In addition, the country's health system faces limitations in addressing these new challenges in prevention and healthcare [9]. Regarding the aging process, a shift in the population pyramid is observed, with projections estimating that the global population aged 60 and older will increase from 900 million in 2015 to more than 1.4 billion by 2030 [10]. In Colombia, the population is estimated to be aging more rapidly: the number of people aged over 65 exceeds that of children under 5 [11].

The concept of precision medicine refers to medical treatments or preventive strategies, based on a comprehensive understanding of the patient's phenotype, genome, and lifestyle [12]. This advancement would not be possible without the technological advances arising from the Human Genome Project. The continued development of this project has enabled the widespread adoption of sequencing technologies and significantly reduced their cost [4]. In Colombia, some steps toward precision medicine have been taken in other departments and regions with distinct ancestral genetic backgrounds, such as those in Antioquia and Chocó [5–8]. Insufficient knowledge of our own genetic variability may hinder participation in the next revolution in healthcare: the implementation of precision medicine [13].

This article describes the identified pathogenic SNPs and variants associated with cardiovascular disease in the population of Caldas, Colombia. The article was written within the framework of the project titled 'Construction of a Precision Medicine Model Based on the Use and Exploitation of Omics Information and Clinical Data,' known as the Caldas Chapter of the ORIGEN-Colombian Genome Project. The project is led by the Centre for Bioinformatics and Computational Biology of Colombia—BIOS and carried out in collaboration with the University of Caldas, the National University of Colombia, the University of Manizales, the Autonomous University of Manizales, the Catholic University of Manizales and University of Quindío.

Results

Population and sample

The project population includes people from the department of Caldas, Colombia. The sample consists of 250 participants divided

into three groups: individuals diagnosed with cardiovascular disease (CVD), a control group, and an older adult group. An analysis of variance (Anova) was performed and statistically significant differences ($P < 0.05$) were found in age between the groups: in the CVD group ($n = 100$), ages ranged from 22 to 73, with a mean of 52.7 years; in the control group ($n = 50$), ages ranged from 22 to 58, with a mean of 43.8 years; and in the older adult group ($n = 100$), ages ranged from 65 to 89, with a mean of 79.4 years. No significant differences in sex distribution were observed ($P > 0.05$).

Clinical characteristics of the CVD group

The analysis was based on clinical records from patients diagnosed with cardiovascular disease, in which the most frequent diagnoses were hypertension (29%), acute myocardial infarction (27%), heart failure (5%), cardiac arrhythmia (4%), ischemic heart disease and coronary artery disease (3%).

Pathogenic variants identified by groups

The number of pathogenic variants identified was significantly higher in the elderly group (60 variants), the CVD group (49 variants), and the control group (16 variants). The pathogenic variants identified for each group are presented in Tables 1–3. Table 1 shows the group with cardiovascular disease. Table 2 shows the elderly group. Table 3 shows the control group. In accordance with the ACMG/AMP guidelines, variants were classified as pathogenic, likely pathogenic, of uncertain significance, likely benign, or benign, based on the information provided by the NCBI. Only variants identified as pathogenic or likely pathogenic are included.

Pathogenic SNP related to cardiovascular disease

The ORIGEN Project identified, for the first time, three pathogenic variants potentially associated with cardiovascular disease (CVD) in individuals from the Caldas region of Colombia. Variants identified in the study population were annotated and cross-referenced using the Cardiovascular Disease Atlas. Two pathogenic variants were confirmed in this database, located in the *HFE* and *PERM1* genes. Additional annotation using ClinVar revealed variants of clinical relevance in the *FBN1* gene. The detailed information on each variant, including chromosomal location and allele change, is presented in Table 4.

Prevalence of pathogenic variant carriers

The prevalence of carriers of pathogenic variants potentially associated with CVD was 13.6%, meaning that 34 of the 250 participants carried these variants. Some participants carried two or three variants. They presented the following allele frequencies: *PERM1* (0.096), *HFE* (0.064), and *FBN1* (0.008). (See Table 5).

Network analysis and functional prediction with bioinformatics analysis

To strengthen the interpretation of variants, other bioinformatics analyses, such as network analysis, were performed using the String platform (<https://string-db.org/>). It was found that *PERM1* regulates the expression of selective target genes *PPARGC1A/B* and *ESRRA/B/G*, with functions in glucose and lipid metabolism, energy transfer, contractile function, muscle mitochondrial biogenesis, and oxidative capacity. *HFE* binds to the transferrin receptor (TFR) and reduces its affinity for iron-loaded transferrin (See Supplementary Figs 1 and 2). A functional prediction was performed using PolyPhen-2, which showed that the *HFE* variant (rs1799945) has a score of 0.755 with a sensitivity of 0.85 and a specificity of 0.92. This indicates a protein prediction with a high

Table 1. Pathogenic SNPs identified in the CVD group.

Gene	dbSNP	Chromosome	ID Clinvar	Ref	Alt	Related to
ABCA4	rs756840095	1	236 100	G	A	Retinal dystrophy
AGXT	rs34116584	2	204 069	C	G	Primary hyperoxaluria, type I
ALPL	rs1057517173	1	1 347 313	G	C	Hypophosphatasia
APC	rs901983934	5	1 355 519	C	A	Familial adenomatous polyposis type 1
ASPM	rs769364943	1	585 449	G	A	Microcephaly type 5, primary, autosomal recessive
ATM	rs587779815	11	2 840 869	G	T	Hereditary cancer-predisposing syndrome
B3GAT3	rs535206047	11	577 885	G	A	Articular chondrodysplasia
B4GALNT1	rs398122382	12	60 524	G	A	Hereditary spastic paraplegia
BTD	rs13078881	3	1900	G	C	Biotinidase deficiency
CASR	rs1057520583	3	379 352	T	G	Familial Hypocalciuric Hypercalcemia
CHD3	rs2545128121	17	1 708 782	G	A	Polymorphism, Reclassified - Adrb1 Polymorphism Revel score: 0.817
COL4A3	rs2106267339	2	2 753 752	G	A	Polymorphism, Reclassified - Adrb1 Polymorphism
CR2	rs1161576814	1	2 917 279	C	T	Immunodeficiency, common variable
CREBBP	rs1596852578	16	694 814	G	A	Rubinstein-Taybi syndrome
DNAH1	rs1268065530	3	2 146 566	C	T	Primary ciliary dyskinesia
FBN1	rs113871094	15	36 082	G	A	Marfan syndrome and cardiovascular system
G6PD	rs1050828	X	10 372	C	G	G6PD deficiency
GATA1	rs104894809	X	10 428	G	A	X-linked thrombocytopenia
GJB2	rs771748289	13	17 000	A	G	Autosomal recessive nonsyndromic hearing loss 1A
GP1BA	rs781541857	17	1 677 212	T	C	Bernard-Soulier syndrome
GP9	rs121918036	3	13 530	A	G	Bernard-Soulier syndrome type C
GRM7	rs1390058794	3	972 739	C	T	Neurodevelopmental disorder
HFE	rs1799945	6	10	C	G	Cholesterol levels - HBP - LDL levels
KCNMA1	rs150678882	10	975 413	G	A	Liang-Wang syndrome
MAST1	rs1969917438	19	1 685 938	T	C	Mega-corpus-callosum syndrome
MMACHC	rs121909209	1	1424	C	T	Thiel-Behnke corneal dystrophy
MSH2	rs1060501993	2	408 465	T	C	Hereditary nonpolyposis colorectal neoplasms
NCOA3	rs142951578	20	1 185 585	C	G	Bilateral sensorineural hearing loss
NEB	rs2552227856	2	3 239 913	G	T	Arthrogryposis multiplex congenita-6
NKX2-6	rs267606914	8	791	A	G	Persistent truncus arteriosus
PDYN	rs267606939	20	18 459	G	A	Spinocerebellar ataxia type 23
PERM1	rs7417106	1	1 320 032	A	G	Atrial fibrillation - HDL
PIGQ	rs200661329	16	453 003	G	A	Epilepsy
PLA2G6	rs1060499764	22	402 192	T	C	Abnormal brain morphology
RAG1	rs104894291	11	13 146	G	A	Histiocytic medullary reticulosis
RASA1	rs1554048066	5	440 233	C	T	Capillary-arteriovenous malformation syndrome
RECQL4	rs772896860	8	952 231	A	G	Baller-Gerold syndrome
SCN1A	rs2105862310	2	1 999 484	G	A	Developmental and epileptic encephalopathy
SCN8A	rs1476532344	12	995 416	G	A	Cognitive impairment with or without cerebellar ataxia
SCUBE3	rs751478115	6	981 660	T	C	Abnormal face shape
SERPINC1	rs121909563	1	18 030	C	T	Hereditary antithrombin deficiency
SETD5	rs864321657	3	219 198	C	T	Intellectual disability-facial dysmorphism syndrome
SLC17A5	rs201284672	6	440 272	A	C	Free sialic acid storage disorder
SMC1A	rs1569358774	X	562 395	T	C	Sufficient evidence for dosage pathogenicity
TBCK	rs62321379	4	225 239	T	C	Hypotonia and childhood retardation
TREM2	rs1765488318	6	1 178 350	C	T	Frontotemporal dementia
TRIOBP	rs375857763	22	1 325 235	C	T	Autosomal recessive nonsyndromic hearing loss 28
VHL	rs786202787	3	186 220	A	G	El síndrome de von Hippel-Lindau
VPS33B	rs11542638	15	587 506	G	A	Arthrogryposis, renal dysfunction, and cholestasis 1

probability of functional damage. Likewise, the prediction was searched for the PERM1 variant—rs7417106 using gnomAD and found to have a maximum score of 0.175 (PolyPhen). This indicates that the probability of functional impairment is much lower (See [Supplementary Fig. 3](#) for details).

Discussion

This study identified three pathogenic variants potentially associated with CVD in the Caldas population. These variants were in the following genes: HFE, PERM1, and FBN1.

The most frequent pathogenic variant across all study groups, located on chromosome 1 (1p36.33), is present in all groups and is related to the peroxisome proliferator-activated receptor gamma coactivator 1 (PGC-1) and estrogen-related receptor (ERR)-induced regulator in muscle 1 (PERM1) gene.

The PERM1 gene plays a role in glucose and lipid metabolism, energy transfer, and muscle contraction. Silencing studies revealed that Perm1 is also required for Pgc1-alpha-induced mitochondrial biogenesis and maximal oxidative capacity. PERM1 mRNA expression was induced by exercise in humans and mice and was reduced in atrophied human muscle with amyotrophic

Table 2. Pathogenic SNPs identified in the senior group.

Gene	dbSNP	Chromosome	ID Clinvar	Ref	Alt	Related to
ABCC2	rs756825916	10	3 013 955	C	T	Dubin-Johnson syndrome
ACADM	rs1570861829	1	643 196	G	A	Medium-chain acyl-coenzyme A dehydrogenase deficiency
ACADS	rs121908003	12	3825	C	T	Deficiency of butyryl-CoA dehydrogenase
AIPL1	rs201883601	17	2 152 271	C	T	Leber congenital amaurosis 4
AVPR2	rs782806507	X	1 203 715	C	T	Diabetes insipidus, nephrogenic, X-linked
BRCA1	rs1555590987	17	548 273	T	A	Breast-ovarian cancer, familial, susceptibility to
BTD	rs13078881	3	1900	G	C	Biotinidase deficiency
BUB1B	rs200884355	15	1 450 254	C	T	Aneuploidy syndrome
CDC45	rs751663397	22	617 938	C	T	Meier-Gorlin syndrome 7
CFTR	rs201124247	7	53 403	A	G	Cystic fibrosis
CHD5	rs760743983	1	995 781	C	T	Seizure
CHEK2	rs1060502710	22	2 584 292	G	C	Familial breast cancer
CIITA	rs567218474	16	664 532	C	T	MHC class II deficiency
CLPB	.	11	1 676 588	C	T	Methylglucosaminiduria
CRYAB	rs104894201	11	16 953	T	C	Myofibrillary myopathy
CTC1	rs200609323	17	529 185	C	A	Dyskeratosis congenita
DDX23	rs2498746301	12	2 664 152	G	A	Congenital bilateral perisylvian syndrome
DICER1	rs775912475	14	933 082	C	T	DICER1-related tumor predisposition
DNAH1	rs1268065530	3	2 146 566	C	T	Ciliary dyskinesia, primary, Spermatogenic failure
ERCC6L2	rs1226402321	9	1 390 409	C	T	Pancytopenia Syndrome
EXTL3	rs554294508	8	417 794	C	T	Musculoskeletal Dysplasia
FANCA	rs199599393	16	974 169	G	C	Fanconi anaemia
FRMD7	rs2520731322	X	1 710 201	A	G	Nystagmus 1, congenital, X-linked
GFPT1	rs2104635880	2	1 070 025	G	A	Congenital myasthenia syndrome
GJB2	rs771748289	13	1 458 348	G	A	Autosomal recessive nonsyndromic hearing loss 1A
GLDC	rs188269735	9	639 955	A	C	Glycine Encephalopathy
HADHA	rs147103714	2	92 594	G	A	Mitochondrial trifunctional protein deficiency
HFE	rs1799945	6	10	C	G	Cholesterol levels - HBP - LDL levels
IBA57	rs1261081427	1	545 651	C	T	Mitochondrial dysfunction
IFT27	rs145338298	22	1 069 785	G	C	Jeune's thoracic dystrophy
IKZF1	rs1131690788	7	428 611	A	G	Pancytopenia
ITGB4	rs121912467	17	14 742	C	T	Junctional epidermolysis bullosa
IVD	rs747273828	15	1 321 422	G	A	Isovaleryl-CoA dehydrogenase deficiency
JAK3	rs2514579231	19	2 719 869	A	G	T-B+ severe combined immunodeficiency due to JAK3 deficiency
KCNJ1	rs104894246	11	9157	G	A	Bartter disease type 2
KCNMA1	rs150678882	10	975 413	G	A	Liang-Wang syndrome
KRAS	rs104894366	12	12 590	G	C	Cardiofaciocutaneous syndrome 2
LEPR	rs144159890	1	631 614	G	A	Obesity due to leptin deficiency
LGR4	rs34804482	11	1 321 211	T	C	Self-limited delayed puberty
MAPKAPK5	rs751979756	12	2 502 375	C	T	Neurocardiofaciodigital syndrome
MITF	rs149617956	3	29 792	G	A	Melanoma, cutaneous malignant, susceptibility to, 8
MTR	rs2528074679	1	2 093 880	A	T	Methylcobalamin deficiency type cblG
MYBPC3	rs397515907	11	42 541	C	T	Hypertrophic cardiomyopathy
NR3C1	rs104893914	5	16 157	C	T	Glucocorticoid resistance
OTOF	rs2465595172	2	2 993 116	A	G	Polymorphism, Reclassified - Adrb1 Polymorphism
PAH	rs62516151	12	583	G	T	Phenylketonuria
PERM1	rs7417106	1	1 320 032	A	G	Atrial fibrillation—HDL
PHEX	rs138773278	X	1 521 884	G	C	Sufficient evidence for dosage pathogenicity
PKD1	.	16	3 338 785	G	C	Polycystic kidney disease, adult type
PNLDC1	rs141903829	6	1 184 273	T	C	Spermatogenic failure
POLR1C	rs191582628	6	30 811	G	A	Treacher Collins syndrome
PURA	.	5	3 255 587	G	C	PURA-related neurodevelopmental disorders
RDH5	rs62638191	12	8003	G	T	Retinal pigmentary dystrophy
RYR1	rs193922868	19	133 061	G	A	Malignant hyperthermia, susceptibility to, 1
SELENBP1	rs1357490520	1	585 260	C	T	Extraoral halitosis due to methanethiol oxidase deficiency
SETX	rs29001665	9	2288	G	A	Spinocerebellar Ataxia
SOX10	rs1555939415	22	692 195	T	C	Waardenburg syndrome type 2A
TSC2	rs397514919	16	64 897	T	C	Hereditary cancer-predisposing syndrome
VDR	rs121909796	12	915 348	C	T	VitD-dependent rickets
WFS1	rs71532862	4	1 373 337	G	A	Optic atrophy

Table 3. Pathogenic SNPs identified in control group.

Gene	dbSNP	Chromosome	ID Clinvar	Ref	Alt	Related to
ACKR1	rs34599082	1	18 396	C	T	Duffy blood group system, FY (bwk) phenotype
ATP1A2	rs2524903070	1	3 234 579	T	C	Not provided
BTD	rs13078881	3	1900	G	C	Biotinidase deficiency
CHD4	rs2540395943	12	975 671	C	T	Sifrim-Hitz-Weiss syndrome
GJB2	rs771748289	13	189 183	G	A	Autosomal recessive nonsyndromic hearing loss 1A
HFE	rs1799945	6	10	C	G	Cholesterol levels - HBP - LDL levels
LAMC2	rs210222351	1	1 374 561	C	G	Gene associated with autosomal recessive phenotype
LEPR	rs144159890	1	631 614	G	A	Obesity due to leptin receptor gene deficiency
MAST1	rs1969917438	19	1 685 938	T	C	Mega-corpora-callosum syndrome
OCA2	rs142988897	15	803 060	C	T	Tyrosinase-positive oculocutaneous albinism
PAH	rs62516151	12	583	G	T	Phenylketonuria
PKD2	rs1324209174	4	562 248	C	T	Polycystic kidney disease 2
RAB3GAP1	rs2468283673	2	3 242 259	G	T	Martolf syndrome
SETD1A	rs2056360507	16	981 933	T	G	Neurodevelopmental disorder
SOX9	rs2143247529	17	2 106 821	T	C	Camptomelic dysplasia
TRIP11	rs1314677608	14	2 056 373	C	T	Achondrogenesis, type IA

Table 4. Pathogenic SNP potentially related to cardiovascular disease.

Number	Gene	Chromosome	Position	SNP	Ref	Alt	Variant type
1	PERM1	1	976 215	rs7417106	A	G	Missense
2	HFE	6	26 090 951	rs1799945	C	G	Missense
3	FBN1	15	48 465 820	rs113871094	G	A	Missense

Table 5. Carriers of pathogenic variants potentially associated with CVD.

Number	Genes and SNP	Control	CVD	Senior	Total	Allele frequency SNP/n
1	FBN1 - rs113871094	1	1	0	2	0.008
2	HFE - rs1799945	2	5	9	16	0.064
3	PERM1 - rs7417106	1	8	15	24	0.096
Total	3 SNP	4	14	24		0.168

lateral sclerosis [14]. In relation to cardiovascular disease, the SNP rs7417106 in the PERM1 gene has been associated with atrial fibrillation and high-density lipoprotein (HDL) levels, according to data from the CVD Atlas [15].

The pathogenic variant that appears second is homeostatic iron regulator gene (HFE). This gene encodes a protein that is primarily expressed on the surface of cells in the liver and intestines. HFE protein is also found in some cells of the immune system [16] and has been associated with hemochromatosis [17]. However, the variant identified with SNP rs1799945, located on chromosome 6 (p22.2), at position 26 090 951, changes a cytosine (C) to guanine (G). This variant is found in the Atlas of CVD and in the GWAS related to CVD, specifically with increased systolic blood pressure, total cholesterol, triglycerides and hypertension [18, 19].

Regarding the two most frequent SNPs identified in this population, the allele frequencies were identified in NCBI's dbSNP or gnomAD. The following were found: In the present study, the allele frequency for PERM1 was 0.096. In other studies, the reference allele ranges from 0.294 in the global population, from 0.060 in the Asian population to 0.507 in the African population, and between 0.189 and 0.322 in the Latin American population.

The allele frequency found for HFE in this population was 0.064. The allele frequency in the global population was 0.864, ranging from 0.970 in Africans, 0.936 in Asians, and 0.892 in Americans. In Latin America, frequencies were found between 0.878 and 0.899.

The Fibrillin 1 gene (FBN1) is located on chromosome 15 (15q21.1). Fibrillin is the main building block of extracellular microfibrils and is widely distributed in connective tissue, both elastic and non-elastic, throughout the body. Since 1991, the FBN1 locus has been mapped to coincide with the region implicated in Marfan syndrome. Subsequent studies have confirmed that mutations in this gene are the primary cause of the disorder [20, 21]. The variant identified in this study, rs113871094 — present in both the CVD and control groups — has been associated with cardiovascular manifestations, particularly thoracic aortic aneurysm and dissection. These manifestations involve structural abnormalities of the aorta, the main artery that transports blood from the heart to the rest of the body [20, 21].

The identification of cardiac genetic markers helps to understand the relationship between genetics and cardiovascular health, that are associated with cardiovascular disorders [22]. Several gene families are involved in the pathophysiology of CVD, including some related to oxidative stress, such as AGT, IL-6,

VCAM-1, E-selectin, M-CSF, CCR2, MMP9, and SCARB1 [23]. However, none of the genes or variants identified in the present study are mentioned.

An increase in the number of pathogenic variants observed in older adults was noted in this study. Although no national references document this trend in Colombia, international studies suggest potential links between aging, polymorphisms, and neurodegenerative diseases [24].

Other studies show that telomere length (TL) is associated with biological aging, which consequently influences the risk of age-related diseases [25]. A link has also been identified between telomere length and some age-related diseases. Additionally, there is scientific evidence that an increase in the number of components of metabolic syndrome is associated with shorter telomere length, oxidative damage at the level of lipids and DNA, and inflammation, as well as with its other components, such as obesity, hyperglycemia, and hypertension [26].

Although validation and segregation analyses would further strengthen the evidence for the pathogenicity of the identified variants, these procedures were beyond the scope and resources of the present study. The current findings are based on high-quality next-generation sequencing (NGS) data and supported by multiple established databases (ClinVar, gnomAD, OMIM, VarSome, and the CVD Atlas). Nevertheless, future studies should aim to validate these variants and elucidate their inheritance patterns within affected families by performing confirmatory Sanger sequencing and familial segregation analyses.

As is common with cross-sectional studies, one of the limitations of this research is the lack of participant follow-up and the absence of a genotype–phenotype relationship. These aspects require further investigation.

These findings represent a novel and significant contribution, as this is the first report of pathogenic variants potentially associated with CVD identified in the Caldas population. This is particularly relevant in the context of Colombia, where genomic research remains limited, and population-specific genetic data are scarce. The underrepresentation of Latin American and specifically Colombian populations in large-scale genomic studies has been repeatedly highlighted as a barrier to equitable implementation of precision medicine. While these findings are promising, several limitations must be acknowledged, including unequal sample sizes across study groups. Furthermore, although the pathogenicity of the variants was supported by established databases (ClinVar, gnomAD, OMIM, VarSome, and the CVD Atlas), functional assays were not performed to confirm their effects in the local population context. Although international efforts have advanced the identification of genetic risk factors for cardiovascular disease, most studies have focused on populations of European or Asian ancestry. In contrast, studies in Latin America are rare, fragmented, or confined to specific case series. This gap reinforces the urgent need for locally generated genomic data to inform risk stratification and therapeutic strategies in diverse populations.

While precision medicine holds great promise, its development in low- and middle-income countries such as Colombia faces challenges that extend beyond economic cost. These include limited access to biotechnological infrastructure, insufficient national genomic databases, and a lack of training in genomics among healthcare professionals [27]. Addressing these structural limitations is essential to ensure that the benefits of genomic science are equitably distributed.

Future studies should address these limitations to strengthen the clinical applicability of genomic findings and to advance

precision medicine strategies tailored to diverse and admixed populations such as those in Colombia.

Materials and methods

Sample collection and DNA extraction

Blood samples were collected following a standardized protocol that included prior informed consent, obtained on-site during participant recruitment. Sample collection was performed by certified healthcare personnel in both clinical settings and mobile field campaigns across the department of Caldas. Genomic DNA was extracted from whole blood using the ReliaPrep™ Blood gDNA Miniprep System (Promega), following the manufacturer's protocol. DNA quality was validated through quantification (Quantus™ Fluorometer), purity (Accuris™ SmartDrop; A260/280 and A260/230), and integrity (1% agarose gel electrophoresis in TBE buffer).

Exome library preparation and sequencing

Exome enrichment was carried out using the Agilent SureSelect Human All Exon v8 library together with Oxford Nanopore Technologies (ONT) sequencing. Recent improvements in ONT chemistry and bioinformatic workflows have increased read accuracy and coverage uniformity, supporting reliable and consistent coverage of targeted exonic regions when paired with the SureSelect v8 capture system [28].

Genomic DNA was enzymatically fragmented, end-repaired, A-tailed, and ligated to molecularly barcoded adapters (SureSelect XT HS2). Captured libraries were then hybridized to biotinylated RNA probes, amplified, purified with AMPure XP beads, and pooled (four samples; 3 µg total DNA). Final libraries were ligated to ONT-specific adapters (SQK-LSK114), quantified, and loaded onto R10.4.1 flow cells for sequencing on the PromethION P2 Solo platform.

Bioinformatic analysis

Exome data from Oxford Nanopore sequencing were processed through a custom bioinformatic pipeline designed for variant detection and interpretation in the context of cardiovascular disease and aging. Raw signal data (POD5) were basecalled using Guppy [29] with the R10.4.1 'sup' model (–config dna_r10.4.1_e8.2_400bps_sup.cfg) and GPU acceleration (–device cuda:all). Basecalled reads were quality-assessed with NanoPlot [30] to summarize read-length distributions and Q-score profiles. (See [Supplementary File 1](#) y [Supplementary File 2](#)).

Reads were then aligned to the GRCh38 human reference genome using Bowtie2 [31] with timing output enabled (–t), the GRCh38 Bowtie2 index specified (–x), single-end FASTQ input (–U), and 28 threads (–p 28); alignments were streamed directly to BAM using Samtools [32] with the reference FASTA provided (samtools view -bT GRCh38.fna).

BAM files were coordinate-sorted with Samtools using 28 threads (samtools sort -@ 28), duplicates were removed during duplicate marking (samtools markdup -r -@ 28), and the final BAMs were indexed (samtools index -@ 28). Variant calling was performed with Longshot v0.4.3 [33] using the processed BAM and GRCh38 reference (longshot -bam <dedup.bam> –ref GRCh38.fna –out <sample>.vcf). Variant callsets were then filtered to retain only records with QUAL > 15 while preserving VCF headers using an awk filter on the QUAL field (\$6 > 15). Structural annotation was performed with snpEff v5.2 [34] using the GRCh38 database build GRCh38.99, verbose mode, and an HTML summary report (snpEff -Xmx200G -v -stats annotated_snpEff.html).

GRCh38.99 filtered_numeric.vcf > annotated_snpeff.vcf). Additional annotation was generated with dbNSFP using the search_dbNSFP43c tool with Java memory allocation (java -Xmx100g -cp search_dbNSFP43c -i filtered_numeric.vcf -o annotated_dbnsfp.vcf -v hg38).

Identified variants were further filtered against cardiovascular gene panels, and those lacking SNP identifiers were manually curated using gene-disease relationship data. The pipeline ensured reproducibility and traceability through implementation on the Origen-Colombian platform. Finally, curated variants were validated and highlighted using the Atlas CVD [35] (<https://ngdc.cncb.ac.cn/cvd/>), which provides clinically relevant associations with cardiovascular disease.

To strengthen the interpretation of variants, other bioinformatics analyses, such as network analysis, were performed using the String platform [36] (<https://string-db.org/>). A functional prediction was also performed using PolyPhen-2 [37] (<http://genetics.bwh.harvard.edu/pph2/>).

The pipeline ensured reproducibility and traceability through implementation on the Origen-Colombian platform.

Analysis of clinical information

The classification was performed by qualified medical personnel. Pathogenic and likely pathogenic variants were retrieved from the VCF files and cross-referenced using SNP identifiers in the Cardiovascular Disease Atlas (CVD Atlas: <https://ngdc.cncb.ac.cn/cvd/>), in order to determine their relevance to known CVD-associated genes. When variants were not present in the CVD Atlas, their clinical significance was further assessed using ClinVar [38] (<https://www.ncbi.nlm.nih.gov/ClinVar/>) and VarSome [39] (<https://varsome.com/>). For variants lacking SNP identifiers or prior classification, additional annotation was performed using OMIM [40] (<https://omim.org/>) and gnomAD [41] (<https://gnomad.broadinstitute.org/>), based on chromosomal position and gene context.

Data systematization and processing

Data systematization and statistical analysis were performed using Excel 2021® (Microsoft) and Statgraphics-Centurion 19-X64® (Statgraphics Technologies). Quantitative variables were described as mean, standard deviation (SD), and 95% confidence intervals (CI), while categorical variables were reported as frequencies and percentages. Associations between variables were analyzed, and statistical significance was defined as $P < 0.05$. The study was approved by the Bioethics Committee of the University of Manizales (Record CBE06–2023).

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Author contributions

Olga Alicia Nieto-Cardenas (Conceptualization, Formal analysis, Investigation, Methodology, Visualization, Writing—original draft, Writing—review & editing), Mariana S Candamil-Cortés (Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Writing—review & editing), Ana Lorena Uribe-Hurtado (Investigation, Methodology, Project administration, Supervision, Writing—review & editing), Edwin David Morales-Álvarez (Conceptualization, Investigation, Methodology, Supervision, Writing—review & editing), Marcela Orjuela-Rodríguez (Conceptualization, Data curation, Investigation, Methodology, Project administration, Validation, Writing—review & editing)

Supplementary data

Supplementary data is available at *Human Molecular Genetics* online.

Conflict of interest statement: The authors declare that there are no conflicts of interest.

References

1. Ministerio de Salud y Protección Social. *Análisis de situación de salud (ASIS) Colombia*, Ministerio de Salud y Protección Social, Bogotá D.C., 2022. <https://www.minsalud.gov.co/sites/rid/Lists/BibliotecaDigital/RIDE/VS/ED/PSP/asis-colombia-2022.pdf>. 2022. Accessed June 2025.
2. BIOS. *Construcción de un modelo de medicina de precisión con base en el uso y aprovechamiento de la información ómica, datos clínicos y paraclínicos mediante el uso de tecnologías 4.0 en el departamento de Caldas*, BIOS, Manizales, 2025, Código BPIN 2021000100368.
3. Fernández-Sanlés A, Sayols-Baixeras S, Subirana I. et al. Association between DNA methylation and coronary heart disease or other atherosclerotic events: a systematic review. *Atherosclerosis*. 2017;**263**:325–333. <https://doi.org/10.1016/j.atherosclerosis.2017.05.022>.
4. Ginsburg GS, Phillips KA. Precision medicine: from science to value. *Health Aff* 2018;**37**:694–701. <https://doi.org/10.1377/hlthaff.2017.1624>.
5. Rojas W, Parra MV, Campo O. et al. Genetic makeup and structure of Colombian populations by means of uniparental and biparental DNA markers. *Am J Phys Anthropol* 2010;**143**:13–20. <https://doi.org/10.1002/ajpa.21270>.
6. De Castro M, Restrepo CM. Genetics and genomic medicine in Colombia. *Mol Genet Genomic Med* 2015;**3**:84–91. <https://doi.org/10.1002/mgg3.139>.
7. Chande AT, Rishishwar L, Ban D. et al. The phenotypic consequences of genetic divergence between admixed Latin American

- populations: Antioquia and Choco, Colombia. *Genome Biol Evol* 2020;**12**:1516–1527. <https://doi.org/10.1093/gbe/evaa154>.
8. Chande AT, Rishishwar L, Conley AB. et al. Ancestry effects on type 2 diabetes genetic risk inference in Hispanic/Latino populations. *BMC Med Genet* 2020;**21**:132. <https://doi.org/10.1186/s12881-020-01068-0>.
 9. Yepes CE, Marín YA. The challenges of health situation analysis in Colombia. *Biomed*. 2018;**38**:162–172. <https://doi.org/10.7705/biomedica.v38i0.3594>.
 10. Comisión Económica para América Latina y el Caribe (CEPAL). Cuarta conferencia regional intergubernamental sobre envejecimiento y derechos de las personas mayores en América Latina y el Caribe, CEPAL, Santiago de Chile, 2020, <https://repositorio.cepal.org/handle/11362/45289>.
 11. Galeano-Marín ME, Zuluaga-Callejas MI, Saldarriaga-Ruiz GJ. Envejecimiento y Covid-19: los debates que impone la pandemia a un mundo envejeciente. Facultad de Ciencias Sociales y Humanas, Universidad de Antioquia, Medellín, 2020.
 12. Ashley EA. Towards precision medicine. *Nat Rev Genet* 2016;**17**:507–522. <https://doi.org/10.1038/nrg.2016.86>.
 13. De los Campos G, Vazquez AI, Hsu S. et al. Complex-trait prediction in the era of big data. *Trends Genet* 2018;**34**:746–754. <https://doi.org/10.1016/j.tig.2018.07.004>.
 14. Cho Y, Hazen BC, Russell AP. et al. Peroxisome proliferator-activated receptor gamma coactivator 1 (PGC-1)- and estrogen-related receptor (err)-induced regulator in muscle 1 (perm1) is a tissue-specific regulator of oxidative capacity in skeletal muscle cells. *J Biol Chem* 2013;**288**:25207–25218. <https://doi.org/10.1074/jbc.M113.489674>.
 15. Wojcik GL, Graff M, Nishimura KK. et al. Genetic analyses of diverse populations improve discovery for complex traits. *Nat*. 2019;**570**:514–518. <https://doi.org/10.1038/s41586-019-1310-4>.
 16. Barton JC, Edwards CQ, Acton RT. HFE gene: structure, function, mutations, and associated iron abnormalities. *Gene*. 2015;**574**:179–192. <https://doi.org/10.1016/j.gene.2015.10.009>.
 17. Feder JN, Gnirke A, Thomas W. et al. A novel MCH class I-like gene is mutated in patients with hereditary haemochromatosis. *Nat Genet* 1996;**13**:399–408. <https://doi.org/10.1038/ng0896-399>.
 18. Evangelou E, Warren HR, Mosen-Ansorena D. et al. Genetic analysis of over 1 million people identifies 535 new loci associated with blood pressure traits. *Nat Genet* 2018;**50**:1412–1425. <https://doi.org/10.1038/s41588-018-0205-x>.
 19. Prins BP, Kuchenbaecker KB, Bao Y. et al. Genome-wide analysis of health-related biomarkers in the UK household longitudinal study reveals novel associations. *Sci Rep* 2017;**7**:11008. <https://doi.org/10.1038/s41598-017-10812-1>.
 20. Schrijver I, Liu W, Francke U. The pathogenicity of the pro1148Ala substitution in the FBN1 gene: causing or predisposing to Marfan syndrome and aortic aneurysm, or clinically innocent? *Hum Genet* 1997;**99**:607–611. <https://doi.org/10.1007/s004390050414>.
 21. Renard M, Francis C, Ghosh R. et al. Clinical validity of genes for heritable thoracic aortic aneurysm and dissection. *J Am Coll Cardiol* 2018;**72**:605–615. <https://doi.org/10.1016/j.jacc.2018.04.089>.
 22. Varghese TP. Genetic markers of cardiovascular disease. *Curr Probl Cardiol* 2024;**49**:102588. <https://doi.org/10.1016/j.cpcardiol.2024.102588>.
 23. Portilla EC, Muñoz W, Sierra CH. Genes y variantes polimórficas asociadas a la enfermedad cardiovascular. *Rev Colomb Cardiol* 2014;**21**:318–326. <https://doi.org/10.1016/j.rccar.2014.09.003>.
 24. López-Cáceres A, Cruz-Sanabria F, Mayorga P. et al. Association between risk polymorphisms for neurodegenerative diseases and cognition in Colombian patients with frontotemporal dementia. *Front Neurol* 2022;**13**:675301. <https://doi.org/10.3389/fneur.2022.675301>.
 25. Rodríguez-Fernández B, Vilor-Tejedor N, Arenaza-Urquijo EM. et al. Genetically predicted telomere length and Alzheimer's disease endophenotypes: a mendelian randomization study, Alzheimer's. *Res Ther* 2022;**14**:167. <https://doi.org/10.1186/s13195-022-01101-9>.
 26. Gavia-García G, Rosado-Pérez J, Arista-Ugalde TL. et al. Telomere length and oxidative stress and its relation with metabolic syndrome components in the aging. *Biology*. 2021;**10**:253. <https://doi.org/10.3390/biology10040253>.
 27. Anaya JM, Herrán M, Pino LE. Challenges and opportunities for precision medicine in developing nations. *Expert Rev Precis Med Drug Dev* 2025;**10**:1–15. <https://doi.org/10.1080/23808993.2025.2505796>.
 28. Oxford Nanopore Technologies, a. Dorado basecaller, <https://github.com/nanoporetech/dorado>. Version 0.3.1, Accessed June 2025.
 29. Oxford Nanopore Technologies, b. Guppy basecalling software, https://community.nanoporetech.com/technical_documents. Version 6.4.6, Accessed June 2025.
 30. De Coster W, D'Hert S, Schultz DT. et al. NanoPack: visualizing and processing long-read sequencing data. *Bioinformatics*. 2018;**34**:2666–2669. <https://doi.org/10.1093/bioinformatics/bty149>.
 31. Li H. Minimap2: pairwise alignment for nucleotide sequences. *Bioinformatics*. 2018;**34**:3094–3100. <https://doi.org/10.1093/bioinformatics/bty191>.
 32. Danecek P, Bonfield JK, Liddle J. et al. Twelve years of SAMtools and BCFtools. *Giga Science*. 2021;**10**:1–14. <https://doi.org/10.1093/gigascience/giab008>.
 33. Edge P, Bansal V. Longshot enables accurate variant calling in diploid genomes from single-molecule long read sequencing. *Nat Commun* 2019;**10**:566. <https://doi.org/10.1038/s41467-019-08460-0>.
 34. Cingolani P, Platts A, Wang LL. et al. A program for annotating and predicting the effects of single nucleotide polymorphisms. *SnPEff Fly* 2012;**6**:80–92. <https://doi.org/10.4161/fly.19695>.
 35. China National Center for Bioinformation. Cardiovascular Disease Atlas (CVD Atlas), <https://ngdc.cncb.ac.cn/cvd/>. Accessed June 2025.
 36. Szklarczyk D, Gable AL, Nastou KC. et al. The STRING database in 2021: customizable protein–protein networks, and functional characterization of user-uploaded gene/measurement sets. *Nucleic Acids Res* 2021;**49**:D605–D612. <https://doi.org/10.1093/nar/gkaa1074>.
 37. Adzhubei I, Jordan DM, Sunyaev SR. Predicting functional effect of human missense mutations using PolyPhen-2. *Curr Protoc Hum Genet* 2013;**76**:1–41. <https://doi.org/10.1002/0471142905.hg0720.s76>.
 38. National Library of Medicine. ClinVar: Public archive of interpretations of clinically relevant variants, <https://www.ncbi.nlm.nih.gov/clinvar/>. Accessed June 3, 2025.
 39. Kopanos C, Tsiolkas V, Kouris A. et al. VarSome: the human genomic variant search engine. *Bioinformatics*. 2019;**35**:1978–1980. <https://doi.org/10.1093/bioinformatics/bty897>.
 40. Johns Hopkins University. OMIM: Online Mendelian Inheritance in Man. <https://omim.org/>. ©1966–2025, Accessed June 3, 2025.
 41. Karczewski KJ, Francioli LC, Tiao G. et al. The mutational constraint spectrum quantified from variation in 141,456 humans. *Nature*. 2020;**581**:434–443. <https://doi.org/10.1038/s41586-020-2308-7>.